

Generation Interconnection System Impact Study Report

For Queue Project Q14-0339

Mount Harmony 138 kV

May 24, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0339. The Interconnection Customer proposes a 1207.5 MW natural gas combined cycle facility interconnecting at the Mount Harmony 138 kV substation in the Maryland area. The request was received on June 08, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$122,804,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Mount Harmony – Dunkirk 138 kV line is overloaded to 551.6% of its 217 MVA emergency rating for the loss of the Quincey – Mount Harmony 138 kV line. The project contributes 100.0% DFAX (1207.5 MW) to this overload.

The Mount Harmony – Quincey 138 kV Line is overloaded to 549.9% of its 218 MVA emergency rating for the loss of the Dunkirk – Mount Harmony 138 kV. The project contributes 100.0% DFAX (1207.5 MW) to this overload.

For the loss of Quincey – Mount Harmony 138 kV, the Dunkirk – El Reno 138 kV line loads to 486.1% of its 196 MVA rating. Project DFAX: 74.6%.

For the loss of Quincey – Mount Harmony 138 kV Line, the University Of Northern Iowa – El Reno 138 kV loads to 254.6% of its 231 MVA rating. Project DFAX: 46.9%.

Priest Bridge – Quincey 138 kV: 209.0% loading (176 MVA) following the outage of the Dunkirk – Mount Harmony 138 kV; DFAX = 31.9%.

Stranger Creek – University Of Northern Iowa 138 kV: 193.1% loading (218 MVA) following the outage of the Shingletown – Riva Road 345 kV; DFAX = 14.0%.

For the loss of Taber – Dixleta 345 kV, the Tazewell – Taber Substation 138 kV loads to 189.9% of its 238 MVA rating. Project DFAX: 5.3%.

For the loss of Taber – Dixleta 345 kV L, the Tazewell – Port Robson 138 kV loads to 188.4% of its 202 MVA rating. Project DFAX: 5.3%.

The Stranger Crk – Dixie Valley 138 kV is overloaded to 181.4% of its 224 MVA emergency rating for the loss of the Shingletown – Riva Road 345 kV Line. The project contributes 14.0% DFAX (168.8 MW) to this overload.

Levitt – Priest Bridge 138 kV: 172.1% loading (201 MVA) following the outage of the Dunkirk – Mount Harmony 138 kV line; DFAX = 31.9%.

For the loss of Glenannan – Taber 345 kV, the Krugville – Beltsville 138 kV loads to 171.7% of its 213 MVA rating. Project DFAX: 6.6%.

The Chesapeake Beach – El Reno 138 kV is overloaded to 170.0% of its 204 MVA emergency rating for the loss of the Quincey – Mount Harmony 138 kV. The project contributes 27.7% DFAX (334.8 MW) to this overload.

The Prairie Wind Farm – Montpelier 138 kV is overloaded to 170.0% of its 203 MVA emergency rating for the loss of the Taber – Glenannan Substation 345 kV line. The project contributes 7.2% DFAX (86.5 MW) to this overload.

Bay Ridge – Dixie Vly 138 kV: 170.0% loading (231 MVA) following the outage of the Riva Rad – Shingleton 345 kV; DFAX = 14.0%.

Sandy Spring – Krugerville 138 kV: 168.4% loading (205 MVA) following the outage of the Glenannan – Taber 345 kV; DFAX = 6.6%.

Quincey – Cheapeake Beach 138 kV: 166.4% loading (201 MVA) following the outage of the Mount Harmony – Quincey 138 kV; DFAX = 27.7%.

Central Avenue – Quincey 138 kV line: 157.6% loading (208 MVA) following the outage of the Dunkirk – Mount Harmony 138 kV; DFAX = 31.7%.

Rush – Colesville 138 kV: 156.1% loading (208 MVA) following the outage of the Darien – Colesville 138 kV L; DFAX = 5.5%.

For the loss of Taber – Dixleta 345 kV line, the Montpelier – Drum 138 kV loads to 141.8% of its 265 MVA rating. Project DFAX: 8.2%.

Wheelabrator Baltimore Refuse – Westport 138 kV L: 134.9% loading (380 MVA) following the outage of the Taber – Dixleta 345 kV; DFAX = 7.8%.

The Quiney – Mitchellville 138 kV L is overloaded to 131.1% of its 192 MVA emergency rating for the loss of the Quincey – Priest Bridge 138 kV Line. The project contributes 23.7% DFAX (286.5 MW) to this overload.

Wilkins – Highgrove 138 kV: 128.5% loading (439 MVA) following the outage of the Taber – Dixleta 345 kV line; DFAX = 7.7%.

Dixleta Substation – North Pole 138 kV #2: 125.0% loading (329 MVA) following the outage of the Taber – Dixleta Substation 345 kV; DFAX = 5.8%.

For the loss of Taber – Dixleta 345 kV Line, the Wilkins – Wheelabrator Baltimore Refuse 138 kV Line loads to 124.7% of its 484 MVA rating. Project DFAX: 7.7%.

The High Rige – North Ple 138 kV is overloaded to 124.5% of its 558 MVA emergency rating for the loss of the Dixleta – Taber Substation 345 kV. The project contributes 9.8% DFAX (118.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mount Harmony – Dunkirk 138 kV line	Rebuild line with higher-capacity conductor	\$6,436,000	100% (\$6,436,000)
Quincey – Mount Harmony 138 kV	Rebuild line with higher-capacity conductor	\$18,562,000	100% (\$18,562,000)
Dunkirk – El Reno 138 kV line	Rebuild line with higher-capacity conductor	\$1,538,000	100% (\$1,538,000)
El Reno – University Of Northern Iowa 138 kV	Rebuild line with higher-capacity conductor	\$12,444,000	100% (\$12,444,000)
Priest Bridge – Quincey 138 kV	Rebuild line with higher-capacity conductor	\$4,651,000	100% (\$4,651,000)
Stranger Creek – University Of Northern Iowa 138 kV	Rebuild line with higher-capacity conductor	\$3,198,000	100% (\$3,198,000)
Taber – Tazewell 138 kV line	Rebuild line with higher-capacity conductor	\$3,909,000	32% (\$1,257,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tazewell – Port Robson 138 kV	Rebuild line with higher-capacity conductor	\$1,919,000	32% (\$617,000)
Stranger Creek Substation – Dixie Valley 138 kV	Rebuild line with higher-capacity conductor	\$2,028,000	100% (\$2,028,000)
Priest Bridge – Levitt 138 kV Line	Rebuild line with higher-capacity conductor	\$2,949,000	100% (\$2,949,000)
Krugerville – Beltsville 138 kV	Rebuild line with higher-capacity conductor	\$1,103,000	44% (\$489,000)
Chesapeake Beach – El Reno 138 kV	Rebuild line with higher-capacity conductor	\$4,576,000	100% (\$4,576,000)
Prairie Wind Farm – Montpelier 138 kV	Rebuild line with higher-capacity conductor	\$2,747,000	50% (\$1,365,000)
Bay Rdge – Dixie Vly 138 kV Line	Rebuild line with higher-capacity conductor	\$3,411,000	100% (\$3,411,000)
Sandy Spring – Krugerville 138 kV	Rebuild line with higher-capacity conductor	\$6,577,000	44% (\$2,917,000)
Quincey – Cheapeake Beach 138 kV	Rebuild line with higher-capacity conductor	\$36,565,000	100% (\$36,565,000)
Central Avenue – Quincey 138 kV	Rebuild line with higher-capacity conductor	\$11,287,000	100% (\$11,287,000)
Rush – Colesville 138 kV	Rebuild line with higher-capacity conductor	\$7,490,000	35% (\$2,630,000)
Montpelier – Drum 138 kV	Rebuild line with higher-capacity conductor	\$9,474,000	17% (\$1,580,000)
Wheelabrator Baltimore Refuse – Westport 138 kV	Rebuild line with higher-capacity conductor	\$567,000	14% (\$79,000)
Quincey – Mitchellville 138 kV	Rebuild line with higher-capacity conductor	\$3,504,000	100% (\$3,504,000)
Wilkens – Highgrove 138 kV Line	Rebuild line with higher-capacity conductor	\$1,055,000	14% (\$148,000)
Dixleta Substation – North Pole 138 kV #2	Rebuild line with higher-capacity conductor	\$979,000	13% (\$127,000)
Wheelabrator Baltimore Refuse – Wilkens 138 kV line	Rebuild line with higher-capacity conductor	\$1,539,000	14% (\$216,000)
High Rige – North Ple 138 kV	Rebuild line with higher-capacity conductor	\$2,208,000	10% (\$230,000)
	Total allocated to project		\$122,804,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.